

Predicting Hotel Booking Cancellations Using Machine Learning: Across Multiple Datasets.

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Abstract—Cancellations in hotel bookings create real Problem and eat into revenue more than most people realize. Hotels constantly deal with the uncertainty of whether a reserved room will actually be used or end up empty at the last minute. The core problem is that traditional methods for guessing cancellation patterns simply are not accurate enough to handle nowadays complex booking data with dozens of influencing factors. In this paper we build and test five machine learning models on three hotel reservation datasets covering Indian hotels and international Inn hotel records using Logistic Regression K Nearest Neighbors Decision Trees Random Forest and a one dimensional CNN. The goal was straightforward train each model on booking history and see which one predicts cancellations most reliably. Accuracy served as our main performance measure alongside precision recall and F1 score to get a fuller picture. Random Forest came out on top scoring between 87 and 90 percent accuracy across all three datasets while CNN and Decision Tree also held up well above 85 percent. The results confirm that machine learning can give hotels a practical tool for forecasting cancellations and among the models tested Random Forest offers the best balance of performance and reliability.

Keywords: Hotel booking cancellation, machine learning, classification, predictive modeling

I. INTRODUCTION

Hotel booking cancellations create serious problems for the hospitality industry beyond just losing room revenue. When a guest cancels at the last minute the hotel rarely finds a replacement booking in time especially during off peak seasons. This leads to empty rooms that could have been sold to someone else and throws off everything from staffing schedules to food and supply ordering. Research shows that cancellation rates in some hotels can climb above 40 percent which directly impacts profitability and operational planning. Understanding why cancellations happen and predicting them ahead of time has become a priority for hotel managers worldwide.

The difficulty lies in the fact that cancellation behavior is influenced by many interconnected variables that are hard to

capture with simple rules. Things like how far in advance the booking was made what type of room was reserved whether the guest has canceled before and even the season all play a role. Most hotels still rely on basic historical averages or gut feelings to estimate cancellation risk which leads to inaccurate forecasts. When predictions are wrong hotels either overbook and anger customers or underbook and lose potential revenue. A data driven solution that learns from past booking patterns could solve this problem much more effectively than traditional approaches.

In this paper we build classification models using five different machine learning algorithms to predict whether a hotel booking will be canceled. We worked with three datasets the first contains over one hundred nineteen thousand booking records from Indian resort hotels with features like lead time room type and customer demographics. The second and third datasets come from international Inn hotel data and include similar booking information with slightly different column structures and record sizes. All five algorithms Logistic Regression K Nearest Neighbors Decision Tree Random Forest and a one dimensional Convolutional Neural Network were trained on the same preprocessing pipeline with categorical encoding and numerical scaling applied consistently. The goal was to compare their performance fairly across all three datasets and identify which model offers the most reliable predictions.

The models were evaluated using accuracy as the primary metric along with precision recall and F1 score to capture a complete picture of classification performance. Random Forest delivered the strongest results reaching 87 percent accuracy on dataset one 90 percent on dataset two and nearly 89 percent on dataset three. K Nearest Neighbors and Decision Tree also performed well landing in the mid 80s across all datasets while Logistic Regression trailed behind with accuracy in the low 80s. The CNN fell somewhere between the tree based models and the simpler classifiers confirming that ensemble methods are better suited for this

type of tabular booking data. Overall the accuracy differences between algorithms were consistent across all three datasets which strengthens confidence in the findings.

This work demonstrates that machine learning classification offers a practical and effective solution for hotel cancellation prediction. Random Forest in particular proves to be a strong candidate for real world deployment given its consistent high performance and resistance to overfitting. The results suggest that hotels can implement these models without needing complex infrastructure or deep learning expertise. Future work could explore additional feature engineering seasonal segmentation and hyperparameter tuning to push accuracy even further. For now the study provides a solid baseline showing that even standard classification algorithms can deliver meaningful improvements over traditional cancellation forecasting methods.

The main contribution of this paper can be summarized as follows. Developing and implementing five machine learning models specifically tailored for hotel booking cancellation prediction across multiple datasets. Comparing the performance of traditional classifiers like Logistic Regression with ensemble methods and neural networks on real world hospitality data. Evaluating models on three distinct datasets to ensure findings are consistent and generalizable rather than dataset specific. Identifying Random Forest as the top performing algorithm with accuracy reaching up to 90 percent across all datasets. Providing a reproducible experimental framework with preprocessing training and evaluation steps that can be adapted for future research in this domain.

The rest of this paper is organized as follows. Section 2 reviews related literature on cancellation prediction and machine learning applications in the hospitality sector. Section 3 describes the datasets in detail including their features sources and the preprocessing steps applied before training. Section 4 presents the experimental setup the implementation of each algorithm and the evaluation metrics used to measure performance. Section 5 discusses the results compares algorithm performance across datasets and analyzes the implications of the findings. Section 6 concludes the paper with a summary of contributions limitations and directions for future research.

II. RELATED WORK

The research paper explores hotel booking cancellation prediction using Naive Bayes, Multi-Layer Perceptron, and XGBoost in a data stream setup [1]. The author focused on the problem that most previous studies depend on static datasets, while hotel booking behavior can change over time because of seasons, market conditions, and customer decisions. The study used a public hotel booking dataset

collected from two Portuguese hotels, which included 32 features related to booking details, customer information, room type, lead time, and cancellation status. To prepare the data, the author applied data cleaning, feature engineering, label encoding, standardization, and time-based splitting to simulate a real streaming environment. The study also tested several windowing methods, such as expanding window, sliding window, forgetting factor, and ADWIN, to update the models over time. XGBoost achieved the best and most stable performance, while MLP and Naive Bayes also worked well with fixed-size window methods. The findings show that using streaming data with machine learning can improve hotel cancellation prediction, especially when the model is updated regularly with new booking data.

The research paper explores how machine learning can be used to predict hotel booking cancellations and support better decision-making in the hospitality industry [2]. The authors used a Kaggle hotel booking dataset containing 36,285 records and 17 features, including booking details, lead time, room type, special requests, parking availability, and cancellation status. The data was first cleaned and prepared by handling missing values, encoding categorical variables, creating new features, and normalizing numerical values. Several machine learning models were tested, including Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and XGBoost. XGBoost achieved the best performance, with 87.6% accuracy, 85.9% precision, 86.4% recall, 86.1% F1-score, and 90.2% ROC-AUC. The study also found that lead time, special requests, room type, booking channel, parking space, and deposit type were important factors affecting cancellation behavior. The findings show that machine learning, especially XGBoost, can help hotels predict cancellations more accurately, improve revenue management, and reduce operational uncertainty.

The research paper explores hotel booking cancellation prediction using the XGBoost machine learning algorithm [3]. The authors focused on the problem of high booking cancellation rates, which can cause revenue loss and inefficient resource utilization in hotels. The study used a publicly available hotel booking demand dataset and prepared the data through cleaning, categorical encoding, normalization, class imbalance handling, and hyperparameter optimization. The proposed model used XGBoost because it can handle complex relationships, reduce overfitting, and perform well with structured booking data. The model was evaluated using several metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The results showed that XGBoost achieved strong and balanced performance, correctly classifying 13,902 non-canceled bookings and 6,974 canceled bookings. The findings show that XGBoost can help hotels predict cancellations more accurately and support better decisions in revenue management, overbooking control, and customer engagement.

This study looks into how deep learning can be used to figure out whether movie reviews are positive or negative. [?] Using 50,000 reviews from Rotten Tomatoes, the authors cleaned and prepared the data before testing several models. They compared XLNet, BERT, LSTM, and CNN-LSTM, with XLNet performing the best at 87.68% accuracy. BERT followed closely behind, and the CNN-LSTM model showed better results than the standard LSTM by blending two different methods of understanding the reviews. The study shows that sentiment analysis can be a helpful tool for getting a general sense of how people feel about movies, though the findings are mostly based on English reviews and might not hold up as well in other languages or specific subject areas.

The research paper explores how adaptive machine learning can improve hotel booking cancellation prediction during uncertain periods such as the COVID-19 pandemic [4]. The authors focused on the problem that models trained on normal historical data may lose accuracy when customer behavior changes suddenly because of crises or market instability. The study used booking data from four Portuguese hotels, including two city hotels and two resort hotels, with more than 670,000 booking records. The authors trained XGBoost models on pre-pandemic data and then tested their performance during the pandemic period. They also used a dynamic sliding-window approach to update the model with more recent data and handle concept drift. The results showed that adding pandemic-period data improved model performance, increasing AUC by up to 5 percentage points. The findings show that continuously updating machine learning models can help hotels predict cancellations more accurately, improve demand forecasting, and make better revenue management decisions during unstable market conditions.

The research paper explores how machine learning can be used to predict hotel booking cancellations for revenue optimization [5]. The authors focused on the problem that unexpected cancellations can reduce hotel revenue, lower room occupancy, and affect operational planning. The study used a hotel reservation dataset with 83,573 booking entries and 9 important features, including market segment, customer type, room type, deposit status, previous cancellations, and special requests. The data was prepared through exploratory data analysis, missing value handling, duplicate removal, outlier treatment, feature engineering, and class imbalance analysis. Several classification models were tested, including Random Forest, XGBoost, LightGBM, Logistic Regression, Decision Tree, Gradient Boosting, SVM, Naive Bayes, and KNN. After hyperparameter tuning and threshold adjustment, XGBoost achieved the best performance with an F2-score of 0.7897. The findings show that XGBoost can help hotels predict cancellations more effectively, improve overbooking policies, apply targeted customer engagement, and support better revenue management decisions.

The research paper [?] examines how they used Machine Learning models in sentiment analysis among many movie reviews and indicating a movie is worth watching or not. They have split their sentences into words using tokenize, also the model check whether it is positive or negative. Authors used both Naïve-Bayes and Linear support vector machine, stochastic gradient descent classifier and Tf-idf. Linear support vector machines have achieved the best results of them with an (91) accuracy.

The research paper explores hotel booking cancellation prediction using autoencoders and statistical dimensionality reduction techniques [6]. The author focused on the problem that feature engineering can improve prediction performance, but it may also add noise, redundancy, and high-dimensional data. The study used a Kaggle hotel reservation dataset containing 36,275 unique bookings and 19 original variables, then created additional engineered features to improve prediction. The main methods included Principal Component Analysis (PCA), Multiple Correspondence Analysis (MCA), and autoencoders for dimensionality reduction. These reduced feature sets were tested using Random Forest, XGBoost, and Multilayer Perceptron models. The results showed that XGBoost achieved the best overall performance with an F1-score of 0.84 when trained directly on the original features. The findings show that dimensionality reduction is not necessary for tree-based models, but autoencoder features can improve neural network performance, making MLP more competitive for hotel cancellation prediction.

The research paper explores hotel booking cancellation prediction by comparing city hotels and resort hotels using machine learning and deep learning models [7]. The author focused on the problem that most previous studies treat hotels as one group, while cancellation behavior may differ between city and resort hotels. The study used a public hotel booking dataset from Portugal with 119,390 observations and 31 variables, including booking details, customer information, market segment, lead time, and cancellation status. The methodology included data cleaning, stratified sampling, feature selection, class imbalance handling, cross-dataset testing, and Optuna hyperparameter tuning. Several models were tested, including Logistic Regression, Random Forest, and TabNet. The results showed that Random Forest achieved the best performance, with an F1-score of 0.85 for city hotels and 0.93 for resort hotels. The findings show that hotel-specific models can perform better for their own hotel type, but combined models may provide more stable performance across both city and resort hotels.

The research paper explores hotel booking cancellation prediction using stacked machine learning and SHAP-based interpretability [8]. The authors focused on the problem that traditional hotel cancellation systems depend on historical trends, manual judgment, and rule-based assumptions, which may fail to capture complex customer behavior. The proposed

framework used several machine learning models, including Logistic Regression, Decision Tree, Random Forest, and Multi-Layer Perceptron. Principal Component Analysis was applied to reduce feature dimensionality and improve computational efficiency. The study also used a stacking classifier to combine the strengths of multiple models and improve prediction accuracy. SHAP was integrated to explain the effect of each feature on cancellation predictions and make the model more transparent for hotel managers. The findings show that the PCA-stacking framework can improve hotel cancellation prediction and support better decisions in revenue optimization, inventory control, and targeted marketing.

The research paper explores hotel room cancellation and no-show prediction using machine learning models [9]. The author focused on the problem that most previous studies predict only general cancellations, while early cancellations, late cancellations, and no-shows have different effects on hotel revenue and room occupancy. The study used a public hotel booking dataset from two Portuguese hotels with 119,390 records and 31 features. The methodology included data cleaning, preprocessing, one-hot encoding, cyclic date transformation, SMOTE for class imbalance, hyperparameter tuning, and permutation importance analysis. Several models were tested, including Random Forest, XGBoost, and Artificial Neural Networks. The results showed that XGBoost achieved the best overall performance, with 0.84 accuracy and 0.61 macro F1-score on the test set, while Random Forest achieved 0.81 accuracy and 0.58 macro F1-score. The findings show that machine learning can help hotels predict early cancellations, late cancellations, and no-shows, but minority classes remain difficult to predict because of class imbalance.

The research paper explores hotel room cancellation prediction using machine learning models [10]. The authors focused on the problem that hotel cancellations can reduce occupancy, waste resources, and negatively affect revenue management, especially when cancellation rates range between 20% and 40%. The study used a hotel booking dataset collected from two Portuguese hotels between 2015 and 2017, including both city hotel and resort hotel records. The methodology included data cleaning, missing value handling, outlier treatment, categorical encoding, normalization, SMOTE balancing, and train-test splitting. Three machine learning models were tested, including Decision Tree, Random Forest, and XGBoost. The results showed that XGBoost achieved the best performance with around 90% accuracy, 90% precision, 90% recall, 0.90 F1-score, and strong AUC-ROC performance. The findings show that machine learning, especially XGBoost, can help hotels predict cancellations more accurately and support better decisions such as dynamic pricing, low-risk customer prioritization, and improved booking management.

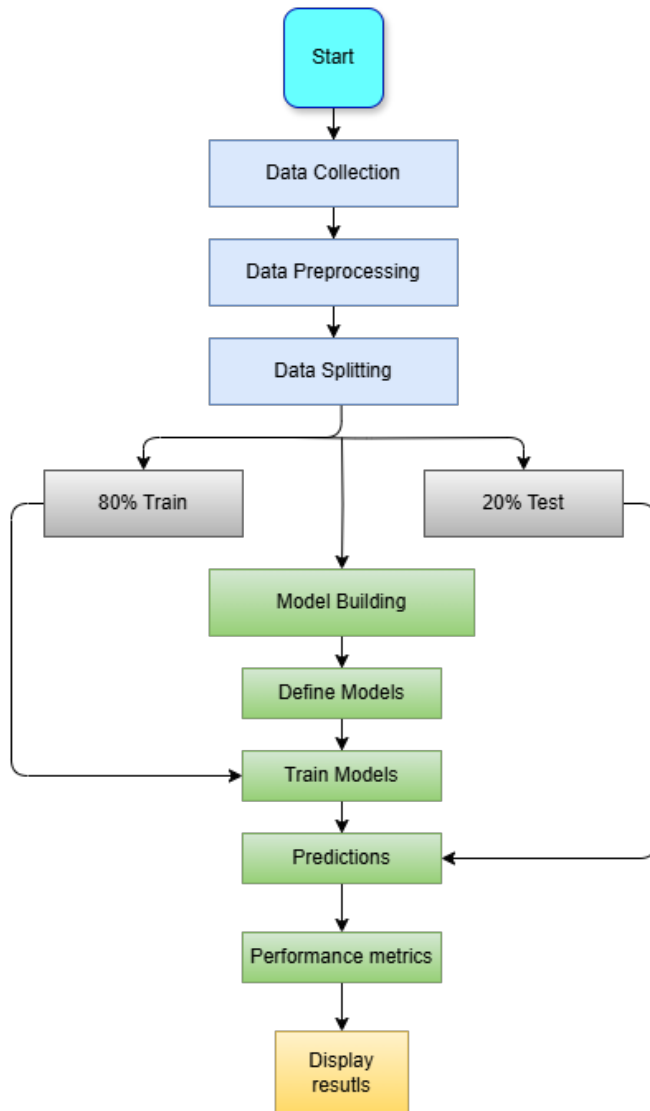
The research paper explores hotel booking cancellation prediction using feature interaction and machine learning models [11]. The authors focused on the problem that many existing models do not fully capture the relationships between booking features, even though feature interactions can strongly affect cancellation behavior. The study used a Kaggle hotel reservation dataset containing 87,005 records with 31 feature variables and one target variable. The methodology included data preprocessing, missing value handling, outlier treatment, categorical encoding, feature extraction, feature interaction generation, and Bayesian probability feature enhancement. Several models were tested, including Logistic Regression with Lasso, Decision Tree, MLP, ANN, Random Forest, XGBoost, LightGBM, and Gradient Boosting. The results showed that Gradient Boosting achieved the best performance, with 93.53% accuracy, 87.67% recall, 92.40% precision, and 89.97% F1-score. The findings show that combining feature interaction with machine learning can improve hotel cancellation prediction and help hotels make better decisions in pricing, resource allocation, and cancellation-risk management.

The research paper explores hotel booking cancellation prediction by comparing machine learning algorithms and neural network models [12]. The author focused on the problem that hotel cancellations can cause major revenue loss, and hotels need accurate prediction models to support overbooking, pricing, and retention strategies. The study used a hotel booking demand dataset from Portugal with 119,390 observations and 32 features, and also tested historical weather forecasting temperature data as an external feature. The methodology included data preprocessing, missing value handling, outlier removal, feature engineering, feature scaling, encoding, feature selection, SMOTE balancing, and hyperparameter tuning. Several models were tested, including Random Forest, Support Vector Machine Ensemble with different kernels, Radial Basis Function Neural Network, and a Dummy baseline model. The results showed that Random Forest achieved the best performance, with 81% F1-score for canceled bookings and 90% F1-score for non-canceled bookings. The findings show that Random Forest is highly effective for hotel cancellation prediction, while SHAP analysis identified lead time, total special requests, deposit type, previous cancellations, and agent information as key cancellation factors.

Ref	Year	Method / Algorithm	Results / Accuracy	Advantages	Disadvantages
[1]	2024	Naive Bayes, MLP, XGBoost in a streaming data setup with expanding window, sliding window and ADWIN.	XGBoost was the most stable and accurate model. MLP and Naive Bayes also worked well with fixed window methods.	Handles concept drift over time. Closer to real hotel booking behavior since data keeps changing.	Complex streaming pipeline. Limited to Portuguese hotel data only. Hard to deploy in practice.
[2]	2025	Logistic Regression, Decision Tree, Random Forest, SVM, KNN and XGBoost on a Kaggle hotel dataset of 36,285 rows.	XGBoost reached 87.6% accuracy, 85.9% precision, 86.4% recall, 86.1% F1 and 90.2% ROC-AUC.	Many models compared. Identified lead time, room type, deposit and parking as key features.	Single dataset only. No class imbalance handling. No deep learning comparison.
[3]	2025	XGBoost alone, with cleaning, categorical encoding, normalization, class balance handling and hyperparameter tuning.	Correctly classified 13,902 non-canceled and 6,974 canceled bookings on the public booking demand data.	Reduces overfitting. Handles complex relations between structured booking features.	Only one algorithm tested. No baseline comparison. No deep learning model included.
[4]	2026	XGBoost with a dynamic sliding window approach trained on pre-pandemic data and tested during the COVID-19 period.	Adding pandemic period data improved AUC by up to 5 percentage points.	Adapts to sudden behavior shifts. Useful during crises and unstable market conditions.	Requires constant retraining. Computationally expensive. Only Portuguese hotels were used.
[5]	2025	Random Forest, XGBoost, LightGBM, Logistic Regression, Decision Tree, Gradient Boosting, SVM, Naive Bayes, KNN with tuning.	XGBoost was the best with an F2 score of 0.7897 after threshold tuning.	Very wide comparison of nine algorithms. Threshold tuning improved minority class recall.	Focus on F2 not accuracy. Single dataset. Business interpretability limited.

Fig. 1: Summary of the reviewed papers on hotel booking cancellation prediction.

III. PROPOSED METHODOLOGY



A. Datasets Descriptions

Three separate hotel booking datasets were used in this study to ensure the findings are not limited to a single data source. The first dataset contains 119,390 booking records from Indian resort hotels spanning cities like Mumbai Delhi Bangalore and Chennai. It includes 33 features covering reservation details such as lead time arrival date room type meal plan customer type and previous cancellation history along with a binary target column labeled canceled. The second dataset comes from international Inn hotel data and contains approximately 36,000 records with 19 features including number of adults weekend and week night stays room type and booking status as the target variable. The third dataset is another Inn hotel variant with similar structure but different column naming conventions and slightly altered feature distributions. All three datasets share common predictors like lead time room type and special requests which are known to influence cancellation behavior. Having multiple datasets allows us to test whether the models generalize well across different hotel types and geographic regions.

B. Data Preprocessing

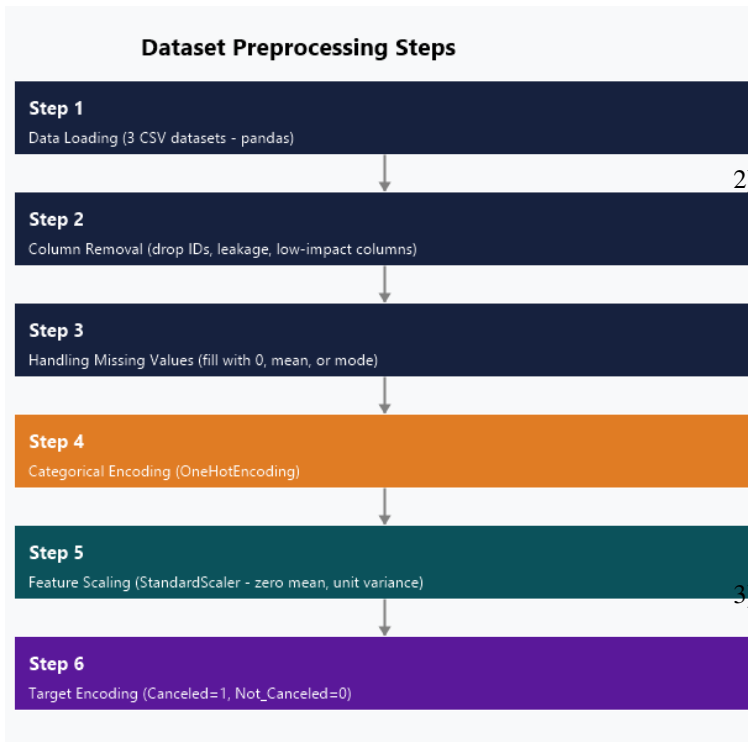


Fig. 2: Preprocessing Steps

Raw data rarely comes ready for machine learning and these datasets were no exception. For the first dataset we dropped several columns that either contained too many missing values or would leak future information about the target. Columns like company and agent had over 85 percent missing values while reservation status and reservation status date directly reflect whether a cancellation happened making them unsuitable for prediction. The children column had only 4 missing values so we filled those with zero and the country column had 488 missing entries which we replaced with the most frequent country. Categorical features such as hotel arrival month meal type market segment and customer type were transformed using one hot encoding to convert them into numerical format. Numerical features like lead time adr and number of special requests were standardized using StandardScaler to give them zero mean and unit variance so models are not biased toward features with larger numeric ranges. For datasets two and three similar preprocessing was applied including handling missing values encoding categorical variables and scaling numerical features. The preprocessing pipeline was kept consistent across all three datasets to ensure fair comparison between models.

1) Dataset 1: Preprocessing Steps:

1) Data Loading:

The first dataset was loaded from a CSV file containing hotel booking records from various Indian cities. It included 119,390 rows and 33 columns covering everything from lead time and room type to customer

demographics and reservation status. The data was read using the pandas library which made it easy to inspect the structure check for missing values and prepare it for the next cleaning steps.

2) Column Removal:

Several columns were removed because they were not useful or would have caused data leakage. The company column had missing values in over 94 percent of rows and the agent column was missing in about 14 percent of rows so both were dropped. Reservation status and reservation status date were removed because they directly indicate whether a cancellation happened which would make the prediction artificially perfect. City arrival date week number and arrival date day of month were also dropped as they added little predictive value compared to other features.

3) Handling Missing Values:

Only a few missing values remained after dropping columns. The children column had just 4 missing entries out of 119,390 records which were filled with zero assuming those bookings had no children. The country column had 488 missing values which were replaced with the most frequently occurring country in the dataset since the country of origin is an important feature and dropping those rows would lose useful information.

4) Categorical Encoding:

Several features in the dataset were categorical and needed to be converted into numbers. Columns like hotel arrival date month meal type market segment distribution channel reserved room type assigned room type deposit type customer type and country were transformed using one hot encoding. This method created binary columns for each category so the machine learning models could understand and use these features during training.

5) Feature Scaling:

Numerical features in the dataset had very different ranges which can cause issues for models like KNN and Logistic Regression. Lead time for example ranged from 0 to 737 days while adr ranged from negative values up to 5400. All numerical features including lead time stays in weekend nights stays in week nights adults children babies is repeated guest previous cancellations previous bookings not canceled booking changes days in waiting list adr required car parking spaces and total of special requests were standardized using StandardScaler to have zero mean and unit variance.

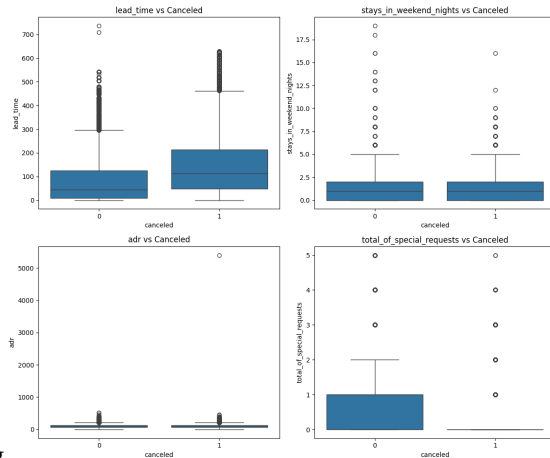
C. Data Visualization

Before training any model it is useful to look at the data with our own eyes. Plots show patterns that summary tables can hide, like which feature seems to push bookings toward cancellation and which feature is roughly the same across both classes. For all three datasets we used the same two types of visualizations to keep things comparable: box plots for the main numeric features grouped by the cancellation label, and grouped bar plots for the main categorical features. The goal here was not to do a deep statistical analysis but to get a feeling for whether the predictors.

1) Dataset 1 Visualization:

1) Sentiment Distribution:

- a) **Numeric Features versus Cancellation:** Fig. 3 shows six numeric features compared between canceled and not canceled bookings in Dataset 1. The lead time plot is the most informative one. Bookings that were canceled have a clearly higher median lead time (around 110 days) compared to bookings that went through (around 40 days). This makes sense in real life because reservations made far in advance are more likely to fall through. plot also shows a useful gap, with non-canceled bookings asking for more special requests on average, which suggests that guests who put effort into their booking are more committed. Stays in weekend and week nights look almost identical between the two classes, so those features are probably weak predictors on their own.



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Fig. 3: Dataset 1 numeric features versus cancellation status.

- b) **Categorical Features versus Cancellation:** Fig. 4 shows nine categorical features broken down by class. The deposit type plot stands out clearly, with non refundable deposits matching cancellations almost perfectly which is a known quirk of this dataset. Market segment also shows a strong

pattern, with online travel agents producing more cancellations than direct or corporate bookings. Meal type and room type show smaller but still visible differences. These plots tell us the categorical features carry real information and that one hot encoding them was worth the cost in dimensionality.

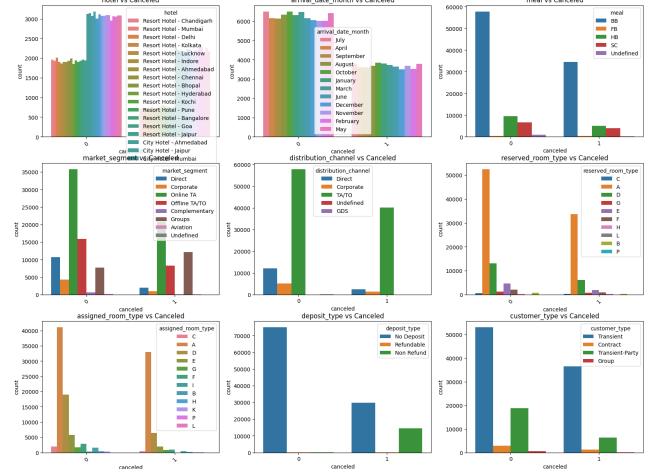


Fig. 4: Dataset 1 categorical features versus cancellation status.

2) Dataset 2 Visualization:

- a) **Numeric Features versus Cancellation:** Fig. 5 shows the same kind of comparison for Dataset 2. The lead time gap is again very visible, with canceled bookings showing a median around 120 days while non canceled sits around 40 days. The pattern is even sharper here than in Dataset 1, which probably explains part of why all models perform better on this dataset. Average price per room is slightly higher for canceled bookings which is interesting because it hints that pricier reservations get canceled more often, maybe because guests are more selective when the cost is higher. Number of weekend nights and number of week nights overlap heavily between the two classes.

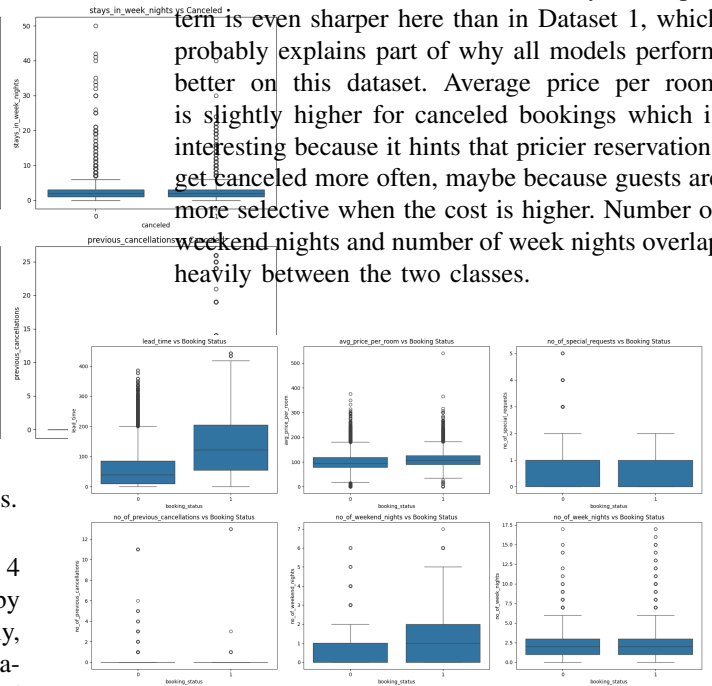


Fig. 5: Dataset 2 numeric features versus booking status.

b) **Categorical Features versus Cancellation:** Fig. 6 compares meal plan, room type and market segment between the two classes. Meal Plan 1 dominates both classes, but the ratio of canceled to non canceled is much higher for Online market segment compared to Offline. Room Type 1 is by far the most common reservation for both canceled and non canceled bookings, while Room Type 4 also shows up frequently. The strong split visible in market segment tells us this feature alone could be a fairly strong predictor in a single feature model.



Fig. 6: Dataset 2 categorical features versus booking status.

3) Dataset 3 Visualization:

1) **Numeric Features versus Cancellation:** Fig. 7 shows eight numeric features for Dataset 3. Lead time once again has the clearest gap, with canceled bookings sitting at a higher median than non canceled. Number of weekend nights also shows a slight upward shift for cancellations which is a small but consistent pattern across datasets. The number of adults and number of children plots are almost flat because most bookings hold exactly two adults and zero children, so these columns add little predictive value. P-C (previous cancellations) is mostly zero for both classes because repeat guests are rare in the data.

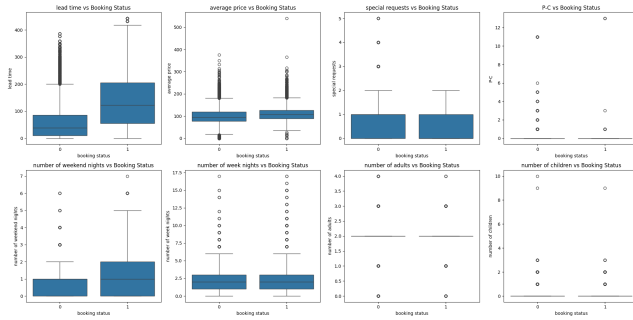


Fig. 7: Dataset 3 numeric features versus booking status.

2) **Categorical Features versus Cancellation:** Fig. 8 shows meal type, room type and market segment in Dataset 3. The shape is almost identical to Dataset 2 which is expected since the two datasets are versions of the same Inn hotel records with renamed columns. The Online segment again accounts for the largest portion of cancellations relative to its size, and Meal Plan 1 plus Room Type 1 remain dominant. The strong consistency between Datasets 2 and 3 supports the idea that our pipeline generalizes.

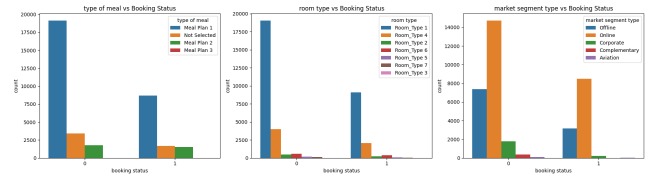


Fig. 8: Dataset 3 categorical features versus booking status.

D. Data Visualization

The goal of this study was to find a model that predicts hotel booking cancellations with high accuracy on tabular data with mixed numeric and categorical features. Five algorithms were chosen on purpose to cover a range of learning styles: a simple linear baseline, a distance based method, a single tree, an ensemble of trees and a neural network. All five were trained on the exact same preprocessed splits described in the previous section so the comparison stays fair. The list below describes each model briefly along with how we applied it to the cancellation prediction task.

- 1: Logistic Regression:** Logistic Regression is one of the most common starting points for binary classification. It draws a linear boundary between the two classes by passing a weighted sum of the features through the sigmoid function which squashes the result between 0 and 1. The number that comes out is read as the probability of cancellation. The reason we kept it in the comparison even though we expected weaker results is that it acts as a sanity baseline. If a fancy model cannot clearly beat plain Logistic Regression then it is probably not worth using.
- 2: K-Nearest Neighbors (KNN):** KNN is a lazy learner, which means it does not really build a model during training. It just stores the whole training set in memory. When a new booking comes in, the algorithm calculates the distance between this booking and every training sample, picks the k closest ones (the default k is 5 in scikit-learn) and assigns the majority class among those neighbors. For our project KNN works because after scaling, similar bookings tend to cluster together. The price we pay is that prediction time grows with the size of the training set, which is something to keep in mind for Dataset 1 with its 95,000 training rows.
- 3: Decision Tree:** A Decision Tree splits the data into smaller and smaller groups by asking yes or no questions on one feature at a time. The split that separates canceled from not canceled the best is chosen at each step using a criterion like Gini impurity. The good thing about trees is they handle numeric and categorical features in the same pipeline and they capture non linear relations, which a linear model cannot. The bad thing is that a single deep tree memorizes the training data and overfits.
- 4: Random Forest:** Random Forest is an ensemble of decision trees. Instead of growing one large tree it grows many smaller trees, each trained on a different random sample of rows and a random subset of features at every split. At prediction time every tree votes and the majority wins. The trick is that the

individual trees overfit in different directions, so their average is more stable than any single tree. This is why Random Forest tends to perform very well on tabular data with a mix of feature types. We used the scikit-learn default settings with `random_state=42`, which builds 100 trees with no depth limit. Random Forest is the model we expected to win going in, based on what other hotel cancellation papers had reported.

5: Convolutional Neural Network (CNN): Convolutional Neural Networks are best known for image and signal tasks, but they can be used on tabular data by treating each row as a one dimensional sequence of features. Our CNN is small and simple: one Conv1D layer with 64 filters and a kernel size of 3, followed by max pooling, flatten, a dense layer of 64 neurons, a dropout layer at 0.3 to reduce overfitting and a final dense layer with a sigmoid output for binary classification. The model was compiled with the adam optimizer and binary cross entropy loss, then trained for 10 epochs with a batch size of 32. We included a deep learning model in the comparison to see whether a neural network can outperform the tree based approaches on this kind of structured booking data, or whether the tabular nature of the problem favors the classical methods.

E. Performance Metrics

Accuracy alone can be misleading for a classification task with imbalanced classes, which is why we used four metrics together to judge model quality. Accuracy gives the overall fraction of correct predictions out of all predictions and is the easiest number to report. Precision tells us, out of all bookings the model flagged as canceled, how many really were. Recall tells us, out of all the bookings that were actually canceled, how many the model managed to catch. F1 score is the harmonic mean of precision and recall and is useful when we want a single balanced number. We also looked at the confusion matrix for each model to see how the errors split between false positives and false negatives. The formulas are listed below.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Confusion Matrix} = \begin{pmatrix} TP & FP \\ FN & TN \end{pmatrix} \quad (5)$$

In our setup TP (True Positive) is a booking that was actually canceled and the model also predicted canceled. TN (True Negative) is a booking that was kept and the model predicted kept. FP (False Positive) is a booking the model flagged as canceled but the guest actually showed up, while

FN (False Negative) is the most damaging case for a hotel, a real cancellation that the model missed. The next section reports these four numbers for each of the five models across all three datasets.

IV. RESULTS AND ANALYSIS

After training each model on the prepared data we collected the results and put them in tables. The five algorithms got the same treatment, same 80/20 split, same random state of 42 and the same preprocessing pipeline, so if one model did better than another it really is because of the model and not because of luck with the data. For every model we looked at four scores: accuracy, precision, recall and F1. All four came from the classification report function in scikit-learn run on the test set after the model finished training. The canceled class is the important one here because the point of the whole project is to catch bookings that will be canceled, not the ones that go through. The first three tables show the results for each dataset on its own and then a fourth table puts the accuracy of all five models side by side so it is easy to compare them at one glance.

Results on Dataset 1 (Indian Resort Hotels, 23,878 test bookings)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8177	0.82	0.82	0.81
K-Nearest Neighbors	0.8241	0.82	0.82	0.82
Decision Tree	0.8298	0.83	0.83	0.83
Random Forest	0.8707	0.87	0.87	0.87
CNN (1D)	0.8526	0.85	0.85	0.85

Fig. 9: Performance of the five models on Dataset 1.

1) Dataset 1 Results Table:

Results on Dataset 2 (Inn Hotel, 7,255 test bookings)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8023	0.80	0.80	0.80
K-Nearest Neighbors	0.8604	0.86	0.86	0.86
Decision Tree	0.8693	0.87	0.87	0.87
Random Forest	0.9009	0.90	0.90	0.90
CNN (1D)	0.8502	0.85	0.85	0.85

Fig. 10: Performance of the five models on Dataset 2.

2) Dataset 2 Results Table:

Results on Dataset 3 (Inn Hotel variant, 7,257 test bookings)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8010	0.80	0.80	0.80
K-Nearest Neighbors	0.8549	0.85	0.85	0.85
Decision Tree	0.8599	0.86	0.86	0.86
Random Forest	0.8871	0.89	0.89	0.89
CNN (1D)	0.8470	0.85	0.85	0.84

Fig. 11: Performance of the five models on Dataset 3.

3) Dataset 3 Results Table:

Accuracy of Every Model Across the Three Datasets

Model	Dataset 1	Dataset 2	Dataset 3
Logistic Regression	0.8177	0.8023	0.8010
K-Nearest Neighbors	0.8241	0.8604	0.8549
Decision Tree	0.8298	0.8693	0.8599
Random Forest	0.8707	0.9009	0.8871
CNN (1D)	0.8526	0.8502	0.8470

Fig. 12: Accuracy of every model across the three datasets.

4) Summary Table:

A. Discussion of Results

Here is a rewritten version in plain text, no LaTeX, no figure refs, more student voice:

The results we got were pretty consistent across the three datasets. Random Forest came out on top every time. On Dataset 1 it scored 87.07 percent, on Dataset 2 it hit 90.09 percent and on Dataset 3 it reached 88.71 percent. The gap between Random Forest and the second best model was around two to four percent each time so it is not just a small lucky difference. Logistic Regression was always at the bottom with accuracy near 80 percent which kind of makes sense since it draws a straight line to split the classes and cancellation behavior is not linear at all. Things like lead time, deposit type and market segment interact in ways that a simple line cannot really pick up. Seeing the same ranking show up three times in a row on different hotel data made us more confident that Random Forest is the right pick and not just a fluke of one dataset.

The CNN was a bit of a letdown. It did worse than Random Forest on all three datasets and even worse than Decision Tree on Datasets 2 and 3, ending up around 84 to 85 percent. We expected more from a neural network but after thinking about

it the reason makes sense. Conv1D layers slide a small filter across the input and assume that features sitting next to each other are related, like pixels in a row of an image. Our data is just a table of columns and the order of those columns does not mean anything. So the CNN was basically picking random groups of features to look at instead of meaningful ones. Tree based models do not have this issue because they pick the single best feature at every split no matter where it sits in the table. This matches what other papers say about deep learning struggling on plain tabular data and our numbers back that up.

KNN and Decision Tree both did much better on Dataset 2 than on Dataset 1. KNN went from 82.41 percent on Dataset 1 to 86.04 percent on Dataset 2. Decision Tree did almost the same thing, climbing from 82.98 percent to 86.93 percent. The reason is probably the size of the feature space. Dataset 1 had 277 columns after one hot encoding which is a lot, and KNN gets confused in high dimensions because distances stop meaning much, while a single Decision Tree gets harder to split cleanly. Dataset 2 only had 28 columns after encoding which is way more manageable. Random Forest did not suffer from this as much because it picks random subsets of features at every split which works as a built in dimensionality fix.

Looking at the confusion matrices below, one thing that stood out was the minority class issue. The non canceled class always had a higher recall than the canceled class on every single model across every dataset. The worst example was Logistic Regression on Dataset 2 which only caught 62 percent of the actual cancellations even though its overall accuracy was 80 percent. Random Forest handled this the best and managed 81 percent recall on canceled bookings in Dataset 2 but still none of the models crossed the 82 percent mark on the minority class. This tells us that even though the overall accuracy looks good, the models are quietly biased toward predicting non canceled bookings. Future work could try using class weights, SMOTE for oversampling, or moving the decision threshold to push the minority class recall up.

B. Confusion Matrix Analysis

The next pages show the confusion matrix for every model on every dataset, fifteen plots in total. Each plot uses the same layout: actual labels on the vertical axis, predicted labels on the horizontal axis. The top left cell counts the bookings correctly predicted as not canceled, the bottom right cell counts the bookings correctly predicted as canceled, and the two off diagonal cells count the two error types. The matrices make it easy to see whether a model leans toward false positives (flagging too many bookings as cancellations) or false negatives (missing real cancellations).

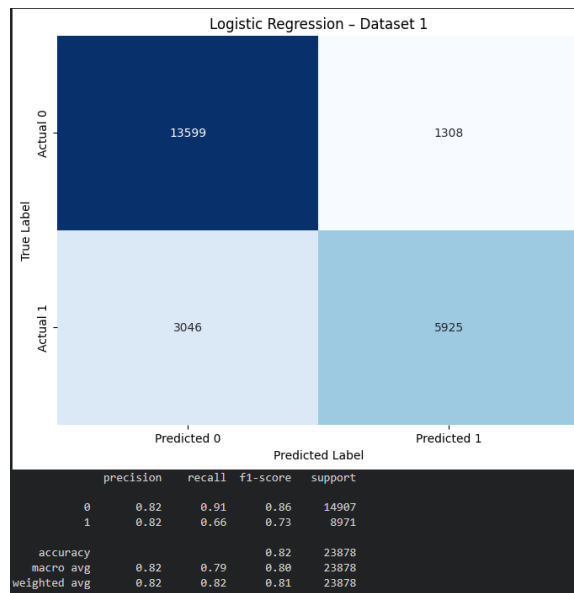


Fig. 13: Confusion matrix of Logistic Regression on Dataset 1.

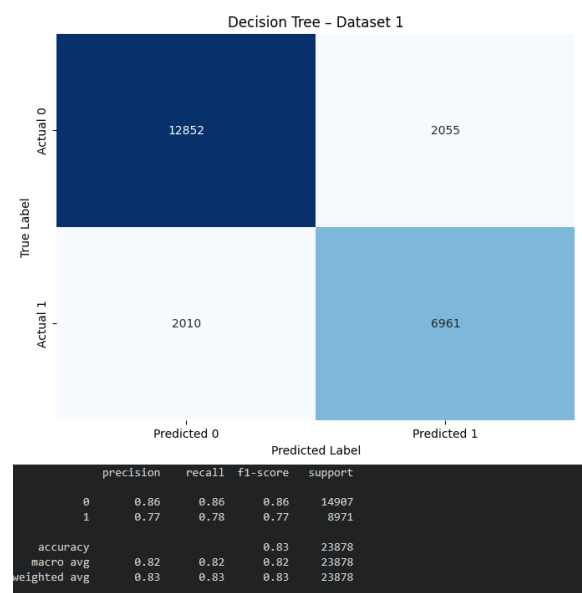


Fig. 15: Confusion matrix of Decision Tree on Dataset 1.

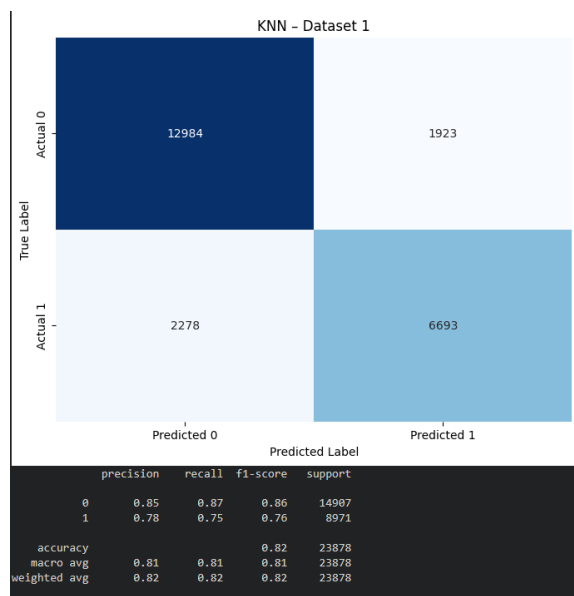


Fig. 14: Confusion matrix of KNN on Dataset 1.

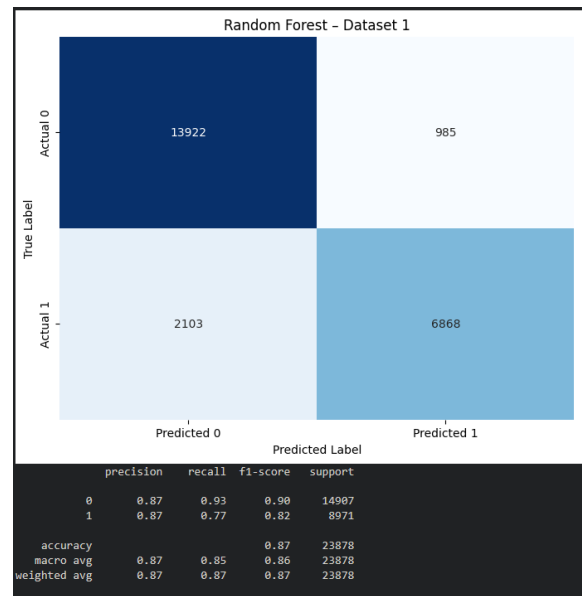


Fig. 16: Confusion matrix of Random Forest on Dataset 1.

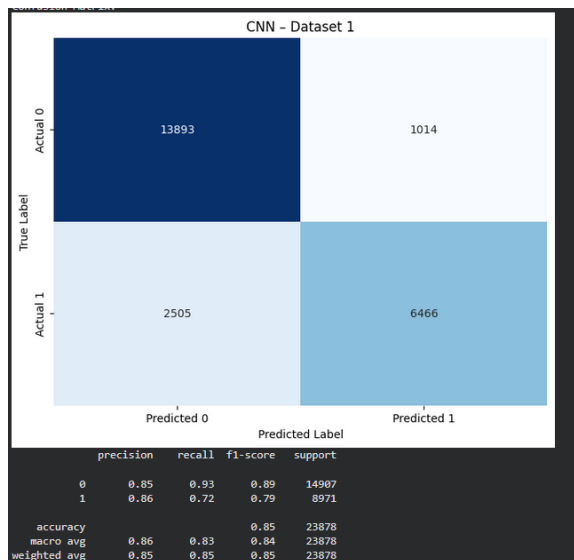


Fig. 17: Confusion matrix of CNN on Dataset 1.

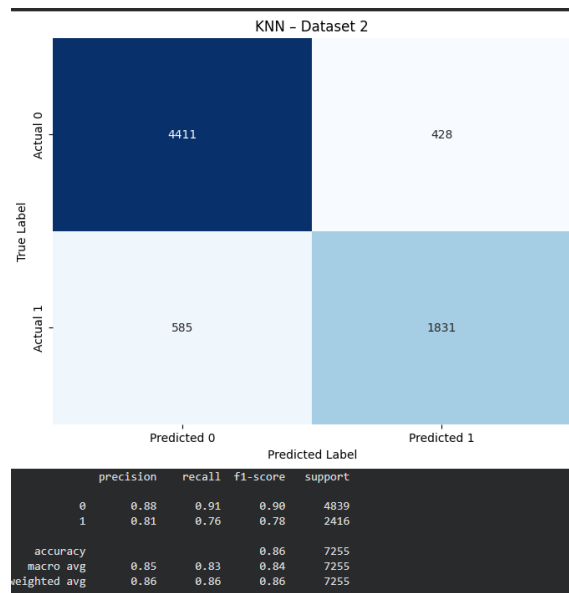


Fig. 19: Confusion matrix of KNN on Dataset 2.

1) *Dataset 1 Confusion Matrices:* The Dataset 1 matrices show that Random Forest produced the smallest number of false positives (985 out of 14,907 non-cancellations) and reached 6,868 true positives out of 8,971 real cancellations. Logistic Regression on the same data missed 3,046 cancellations, which is the worst miss rate among the five models. The CNN did decent work on the negative class but missed 2,505 cancellations, slightly worse than Random Forest.

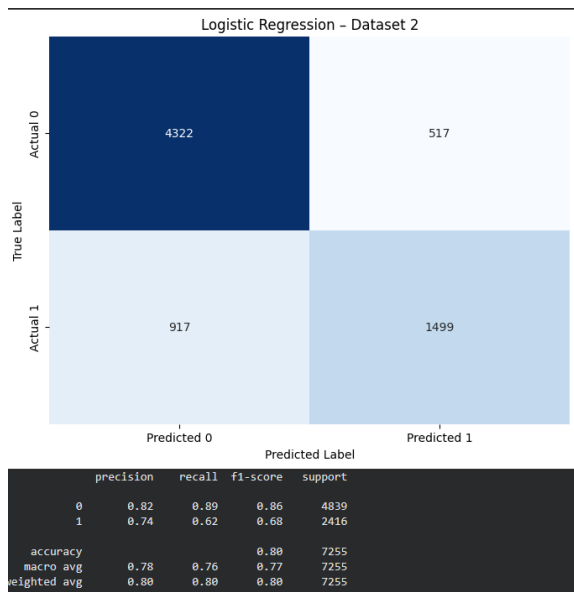


Fig. 18: Confusion matrix of Logistic Regression on Dataset 2.



Fig. 20: Confusion matrix of Decision Tree on Dataset 2.

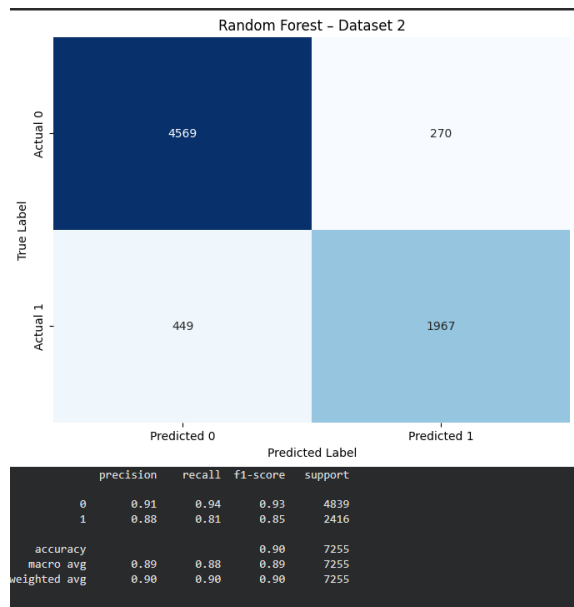


Fig. 21: Confusion matrix of Random Forest on Dataset 2.

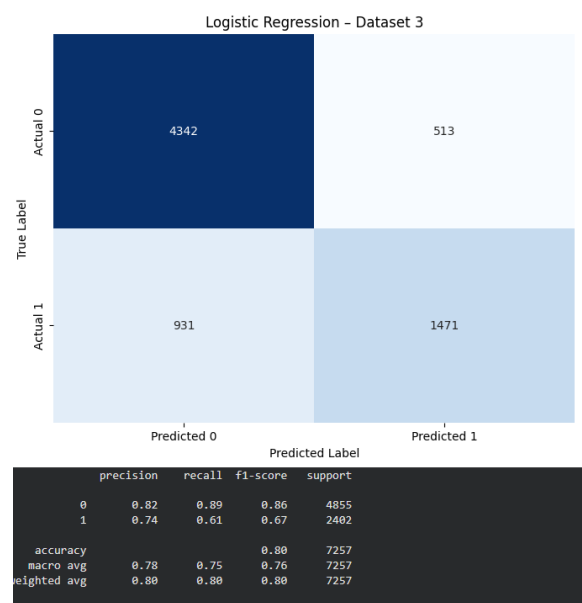


Fig. 23: Confusion matrix of Logistic Regression on Dataset 3.

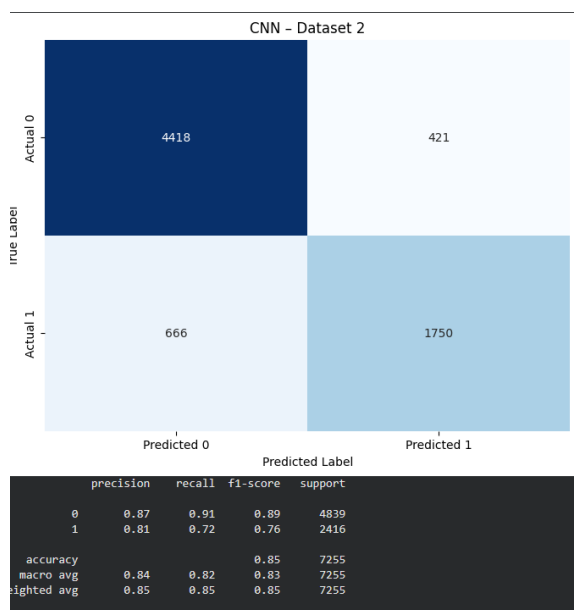


Fig. 22: Confusion matrix of CNN on Dataset 2.

2) *Dataset 2 Confusion Matrices:* Dataset 2 shows the sharpest performance from Random Forest of any dataset. Only 270 not canceled bookings were misclassified as canceled and 449 real cancellations were missed, giving the model 1,967 correct positive predictions. Decision Tree is interesting here because it has perfectly balanced error counts (474 false positives and 474 false negatives), which is rare. Logistic Regression once again struggled with the minority class, missing 917 of the 2,416 cancellations.



Fig. 24: Confusion matrix of KNN on Dataset 3.

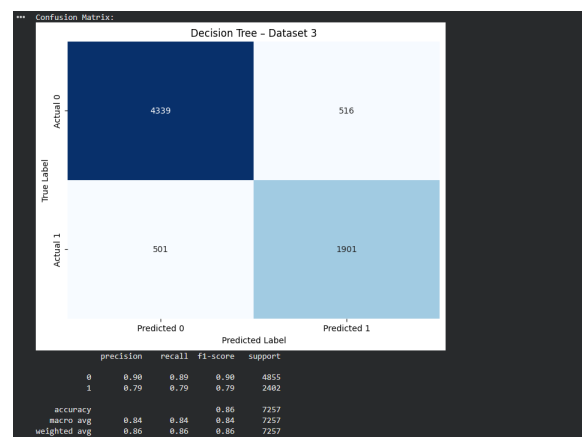


Fig. 25: Confusion matrix of Decision Tree on Dataset 3.

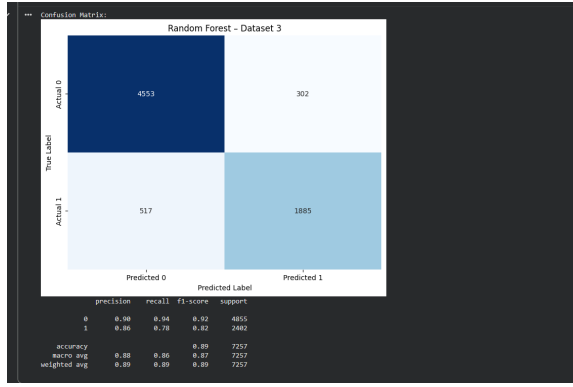


Fig. 26: Confusion matrix of Random Forest on Dataset 3.

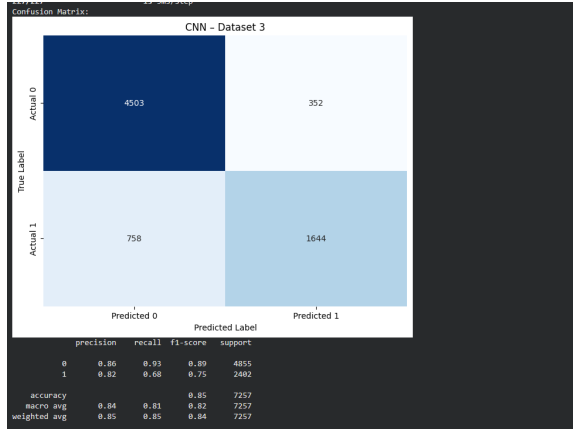


Fig. 27: Confusion matrix of CNN on Dataset 3.

3) *Dataset 3 Confusion Matrices:* Dataset 3 follows the same pattern as Dataset 2 with very similar numbers, which is expected because the two datasets share the same Inn hotel source with slight column renaming. Random Forest again leads with 4,553 correct negatives and 1,885 correct positives. The CNN missed 758 real cancellations here, which is the highest false negative count among the four non linear models. Logistic Regression caught only 1,471 of the 2,402 actual cancellations, confirming that linear models are a poor fit for this problem.

C. Cross Dataset Comparison

When the same algorithm gives similar relative rankings on three independent datasets, the result is more trustworthy than if it happened on only one. In our experiments the order of the five models from best to worst is identical on all three datasets: Random Forest, then either CNN or Decision Tree depending on the dataset, then KNN, with Logistic Regression at the bottom. The accuracy gap between Random Forest and Logistic Regression is around 5 to 9 percentage points depending on the dataset, which is a meaningful difference in real hotel operations.

The strongest performer overall is Random Forest on Dataset 2 at 90.09% accuracy and 0.85 F1 on the canceled class. The weakest performer is Logistic Regression on Dataset

3 at 80.10% accuracy and 0.67 F1 on the canceled class. The fact that even the weakest model still gets 80% right out of the box says that the cancellation problem is reasonably predictable from the features we already have, and that the difference between the models is really about how well each one captures the non linear interactions between lead time, deposit type, market segment and special requests.

For real deployment the recommendation is clear. Random Forest should be the production model because it has the highest accuracy on all three datasets, the best F1 on the canceled class, and the fewest false negatives in absolute count. If interpretability is important to hotel managers, Decision Tree is a reasonable second choice because it can be visualized and explained one split at a time. The CNN and Logistic Regression are worth keeping as references but they are not strong candidates for production in this domain.

V. CONCLUSION

This project focused on predicting hotel booking cancellations using machine learning models. The main idea was to help hotels know which reservations are likely to be cancelled before the guest arrival date. This can help hotels reduce losses, manage room availability better, and make smarter decisions about pricing and overbooking. After cleaning and preparing the dataset, different models were tested, such as Logistic Regression, KNN, Decision Tree, Random Forest, and CNN. The results showed that Random Forest gave the best performance, with accuracy around 87/100, 90/100, and 89/100 across the datasets. This means Random Forest was the most reliable model in this project for learning booking patterns and predicting cancellations. Overall, the project shows that machine learning can be a useful tool for hotels to reduce uncertainty and improve reservation management.

A. Future Work:

There are a few things we want to try if we get more time on this project. The first one is fixing the class imbalance. The cancellation class is the minority class and even our best model missed around 20 percent of real cancellations. Using SMOTE to oversample the canceled bookings or setting the class weight parameter to balanced inside scikit-learn might push the recall on cancellations higher without hurting the other class too much. The second thing is we used default hyperparameters for every algorithm which is not ideal. Running a grid search or a randomized search on Random Forest, KNN and the Decision Tree would probably squeeze out a couple of extra percentage points and make the results stronger.

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